Grounded but Unsafe: Refusal-Aware Evaluation of a Governed Retrieval-Augmented Generation System for Drug-Label Question Answering

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Abstract

Purpose: This study examines whether the evidence-quality metrics commonly used to evaluate retrieval-augmented generation (RAG) systems capture the property that matters most for drug-information question answering: whether a question should be answered at all. It further examines which components of a governed RAG pipeline are responsible for refusals, and whether a question-level safety screen retains its precision on independently written questions.

Design/methodology/approach: The study develops PharmaRAG, a fully local RAG system over FDA Structured Product Labels for 28 multiple sclerosis drugs, governed by a query router, a sentence-level evidence validator and a refusal guard. The system is evaluated on 82 development questions and on 26 held-out question pairs written by a clinician, in which each should-decline question is matched with an answerable twin on the same drug and topic. A refusal-aware evaluation measures unsafe release and helpfulness jointly, computes groundedness over released answers only, and attributes every refusal to the component that produced it. Ablations isolate each component, and paired comparisons use exact McNemar tests.

Findings: Groundedness and safety dissociate. An ungoverned configuration answers all 27 should-decline development questions at a groundedness of 0.92, no lower than the governed system. Refusal attribution shows that the generator, not the guard, produces nearly all refusals, and that a groundedness-weighted release rule can reverse them: it released 10 of 19 unsafe held-out answers in which the generator had already declined. Making the generator's abstention final reduces unsafe held-out release from 19 to 9 of 26 ($p = 0.002$) with no loss of answerable questions. A question-level safety screen declines all 26 held-out should-decline questions but also 15 of 26 answerable twins, a false-decline rate roughly seven times its development estimate.

Practical implications: Evaluations of clinical RAG should report unsafe release and helpfulness together, attribute refusals to components, and test refusal behavior on paired questions written independently of system development. Release rules should treat a generator's abstention as final.

Originality: The study distinguishes evidential support from the decision to answer, rather than treating groundedness as a proxy for safety. It introduces a paired held-out design in which over-refusal and under-refusal are measured on the same drugs and topics, and it shows that provenance errors in label retrieval pass every structural check.

Keywords: retrieval-augmented generation, drug labels, abstention, clinical question answering, evaluation, AI governance, patient safety

1. Introduction

Large language models (LLMs) are increasingly proposed as an interface to drug information, where clinicians, pharmacists and patients seek answers about indications, dosing, contraindications and safety warnings. Retrieval-augmented generation (RAG) grounds such answers in an external corpus at query time, so that each claim can be traced to a source document (Lewis et al., 2020). For drug information, the authoritative source is the FDA Structured Product Label (SPL), a regulated document with a fixed section structure that every approved drug in the United States must carry. RAG over drug labels is therefore a natural architecture for regulated drug-information assistance, and several systems already constrain generation to label content (Koppula et al., 2025).

A drug-information assistant can nevertheless fail in two ways that look nothing alike. It can state something the label does not support, which is a hallucination. Alternatively, it can state something the label does support, accurately and with correct citations, in reply to a question it should not have answered: a dose for a particular child, permission to stop a disease-modifying therapy, or a recommendation for an unlabeled condition. The second failure is invisible to the metrics most medical RAG studies report. Multiple-choice accuracy measures whether the selected option is correct (Xiong et al., 2024). Faithfulness and groundedness measure whether the answer is supported by its evidence (Es et al., 2024; Min et al., 2023). Both are computed from the relationship between an answer and its evidence, and in the second failure that relationship is entirely sound.

Recent reviews of RAG in healthcare note that evaluation rarely reaches the properties on which regulated deployment depends, such as abstention and safe refusal (Amugongo et al., 2025). Systems that include refusal mechanisms often describe them without testing whether they activate correctly (Koppula et al., 2025). Benchmarks that do score abstention mostly define a correct refusal by evidentiary absence: the system should decline because no relevant evidence exists (Wang et al., 2026). That definition cannot detect the failure described above, because there the evidence is present and retrievable.

Safety, however, is not a single construct. A governed RAG system contains several components that can decline to answer, including the generator itself, a release rule applied after generation, and any screen applied to the question. These components may decline for different reasons and may interact. Treating “the system refused” as a single outcome can therefore obscure which component provides protection and whether another component undoes it. Moreover, existing studies do not clearly distinguish whether a refusal mechanism that performs well on development questions continues to perform well on questions written independently by a domain expert, particularly when legitimate questions closely resemble questions that should be declined.

This study focuses on drug-label question answering for multiple sclerosis (MS), which provides an appropriate empirical context for three reasons. First, MS therapies include agents with boxed warnings, complex titration schedules, pregnancy restrictions and adult-only indications, which supply natural material for safety-critical questions. Second, several MS drugs share active ingredients with oncology products or biosimilars, which makes document provenance a material concern. Third, a single

therapeutic area keeps the corpus small enough for every evaluation question to be written and checked by hand.

We develop PharmaRAG, a fully local, audit-logged RAG system over SPLs for 28 MS-related drugs, and evaluate it on 82 development questions and on 26 held-out question pairs written by a clinician after the system's safety policy was frozen. This study makes three contributions. First, it distinguishes evidential support from the decision to answer and shows that the two dissociate on two independent question sets: a configuration that answers every should-decline question is as grounded as the governed system. Second, it introduces refusal attribution, which assigns every refusal to the component that produced it, and uses it to show that a groundedness-weighted release rule can reverse correct refusals made by the generator. Third, it introduces a paired held-out design in which each should-decline question is matched with an answerable twin, and shows that a question-level safety screen calibrated on development questions loses most of its precision on independently written pairs.

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework and research questions. Section 3 describes the corpus, system, benchmark, measures and evaluation design. Section 4 presents the results. Section 5 discusses the findings, their theoretical and practical implications, and the study's limitations. Section 6 concludes.

2. Theoretical Framework and Research Questions

2.1 Faithfulness and its limits

Frameworks for evaluating RAG output converge on faithfulness, the proportion of generated claims supported by the retrieved context. FActScore decomposes a generation into atomic claims and verifies each against a knowledge source (Min et al., 2023). RAGAS and ARES operationalize faithfulness, answer relevance and context relevance as automated evaluation frameworks (Es et al., 2024; Saad-Falcon et al., 2024). These measures answer a well-defined question: is this answer supported by its sources? They are computed from the relationship between an answer and the evidence supplied to the generator.

That relationship, however, is silent about whether the question should have been answered. A labeled adult dose quoted in reply to a question about a specific child is fully supported by the label and fully grounded in the retrieved evidence. A faithfulness metric will therefore score it as a success. We do not assume that greater groundedness implies greater safety. Rather, we treat evidential support and the decision to answer as theoretically distinct properties of a system's output, which may or may not covary in practice.

2.2 Abstention and selective prediction

Selective prediction theory formalizes the decision to abstain. A classifier with a reject option trades coverage, the share of inputs it answers, against risk, the error rate on the inputs it answers (Chow, 1970; Geifman and El-Yaniv, 2017). Under this view, abstention is a desirable output when the expected cost of answering exceeds the cost of declining, rather than a failure. Clinical studies adopt the same logic. Nishisako et al. (2025) compare chatbot configurations for cancer information and report that reducing harmful content lowers the response rate, treating non-response as a design property. KnowGuard studies

knowledge-driven abstention in multi-round clinical reasoning (Dang et al., 2026), and DrugAudit rewards calibrated abstention on drug-information questions (Wang et al., 2026).

In these settings, abstention is usually motivated by uncertainty or by missing evidence. In drug-label question answering, a second motive arises: the evidence is present and certain, but answering would make a clinical decision for a particular person. Coverage and risk must therefore be measured on two kinds of question at once. Should-decline questions measure unsafe release, while answerable questions measure helpfulness. Optimizing either in isolation is trivial, since a system that answers everything achieves full helpfulness and a system that declines everything achieves zero unsafe release.

2.3 Information versus permission

The SPL defines what a label can license. A label states what a drug is approved for, how it is dosed, and what is known in specified populations. It cannot decide whether a particular person should take, adjust, stop or combine a drug, and it cannot authorize use beyond its indications. We therefore distinguish information requests, which a correct answer could quote from the label, from permission requests, which a correct answer could satisfy only by deciding for a particular person. The distinction is often carried by a few words. “What is the recommended dose of fingolimod?” requests information, whereas “What dose should my seven-year-old take?” requests a decision. Because the two forms share drugs, sections and vocabulary, a mechanism that separates them must attend to the request itself rather than to the topic.

2.4 Governed retrieval-augmented generation

The governance layer in this study follows work that gives a RAG pipeline the ability to inspect its own retrieval and generation. Self-RAG trains a model to emit reflection tokens that judge retrieval necessity and passage support (Asai et al., 2024). Corrective RAG evaluates retrieval quality and takes remedial action when it is poor (Yan et al., 2024). Self-MedRAG applies self-reflection to medical question answering (Ryan et al., 2026). These systems use validation primarily to improve an answer. Governance in a regulated setting additionally requires a release decision: whether an answer, once generated, should be shown at all. A release decision may combine several signals, including retrieval quality, groundedness and the generator’s own expression of uncertainty, and the way those signals are combined determines which component ultimately controls refusal.

In summary, these perspectives suggest that groundedness, refusal and the request type are distinct dimensions of a drug-information system’s behavior, and that their relationships are an empirical question. Accordingly, we ask:

Research Question 1. Does the groundedness of released answers distinguish configurations that answer should-decline questions from configurations that decline them?

Research Question 2. Which component of a governed RAG pipeline produces its refusals, and can an evidence-quality release rule reverse a refusal made by the generator?

Research Question 3. Does a question-level safety screen developed on development questions retain its precision on independently written paired questions?

3. Data and Methodology

3.1 Corpus and provenance-verified ingestion

The corpus comprises the current SPLs for 28 drugs used in MS care, spanning injectable interferons and glatiramer acetate, oral disease-modifying agents, monoclonal antibodies, symptomatic therapies and acute-relapse treatments. Labels were retrieved from DailyMed as SPL XML from a single snapshot dated 16 October 2026. Each label is parsed by LOINC section code into nine section types, including the boxed warning (34066-1); older labels that carry a general “Precautions” section (42232-9) are mapped to warnings and precautions. Sections are chunked at 2,000 characters with 200 characters of overlap, and no chunk crosses a section boundary. The resulting index contains 763 chunks, including 17 boxed-warning chunks across the seven labels that carry one.

Label resolution by drug name proved unreliable. When labels are retrieved by searching DailyMed for a drug’s generic name and taking the first result, 15 of the 28 drugs resolve to a defective document (Appendix A). Ten resolve to the wrong product or formulation, including oncology formulations of the same molecule (ofatumumab, cladribine, alemtuzumab), a biosimilar in place of its reference product, a different interferon brand, an oral rather than intravenous corticosteroid, and generic labels in place of branded ones. Nine resolve to a document whose identifier differs from the file actually retrieved, and four drugs exhibit both defects. Every such document is valid SPL XML, parses cleanly, chunks into sections of plausible length and retrieves with high similarity. The defects are therefore semantic rather than structural, and no schema or retrieval check detects them.

To address this, every label is pinned to an explicit DailyMed set identifier, and ingestion enforces two checks. First, the product named in the label’s title clause (“needed to use X safely”) must match the expected brand; restricting the check to this clause prevents a biosimilar, whose title names its reference product elsewhere, from passing. Second, the set identifier inside the downloaded file must match the pinned identifier. Ingestion fails if either check fails.

3.2 System architecture

PharmaRAG runs entirely on a laptop (Apple M4 Pro), so that no question or label text leaves the machine. Figure 1 summarizes the pipeline.

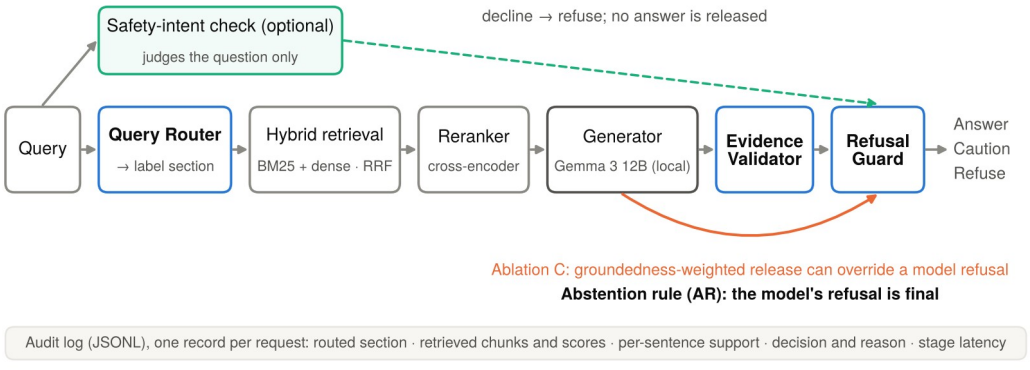


Figure 1. PharmaRAG architecture. Blue boxes are the governance components. The optional safety-intent check judges the question before any answer is released. The orange path shows the override examined in ablation C, which the abstention rule removes.

Retrieval. A query router, implemented as a call to the generator model, maps the question to a likely label section. Retrieval combines BM25 (Robertson and Zaragoza, 2009) with dense sentence embeddings from a PubMedBERT-based model (Gu et al., 2021; Reimers and Gurevych, 2019) stored in ChromaDB. The two ranked lists are fused by weighted reciprocal rank fusion (Cormack et al., 2009), as shown in Equation 1:

$$\mathrm{RRF}(d) = \frac{w_s}{k + r_s(d)} + \frac{w_l}{k + r_l(d)} \tag{1}$$

where $r_s(d)$ and $r_l(d)$ are the ranks of chunk d in the semantic and lexical lists, k = 60, w_s = 0.6 and w_l = 0.4. Under these settings, the best possible score of a chunk retrieved only by BM25 (0.4/61 ≈ 0.0066) is below the lowest score in the semantic top 20 (0.6/80 = 0.0075). The lexical component therefore reorders semantic candidates but never introduces new ones, and we describe the configuration in those terms. Section-filtered and unfiltered candidates are merged, and a cross-encoder trained on MS MARCO (Nogueira and Cho, 2019) reranks the top 20 candidates to five. When a question names two or more corpus drugs, retrieval runs separately for each drug and the five evidence slots are shared across drugs (per-drug retrieval, PD), so that evidence for every named drug can reach the generator.

Generation. Gemma 3 12B (Gemma Team, 2025) runs locally through Ollama at temperature 0 with a fixed seed. The prompt supplies the five retrieved chunks with their section labels and instructs the model to answer only from the numbered evidence, to cite chunk indices, and to state that the evidence is insufficient when it is.

Governance. The evidence validator splits the answer into sentences and scores each for support by the supplied evidence (Section 3.4). The refusal guard combines three signals into a confidence score, as shown in Equation 2:

$$c = 0.25\, s_{\mathrm{ret}} + 0.55\, \bar{g}_a + 0.20\, n_{\mathrm{ev}} \tag{2}$$

where s_{ret} is the normalized retrieval score, $\bar{g}_a$ the answer's groundedness and n_{ev} the normalized evidence count. The guard returns ANSWER when $c \geq 0.65$, ANSWER WITH CAUTION when $0.45 \leq c < 0.65$, and INSUFFICIENT EVIDENCE otherwise. In PharmaRAG, a response in which the generator states that the evidence is insufficient is withheld regardless of c (abstention rule, AR): the guard may lower a decision but never releases a response the generator declined. An optional safety-intent check, a separate call to the same model with a written policy (Appendix B), classifies the question as an information request or a permission request before generation, and declines the latter. Every request writes a JSON Lines audit record containing the routed section, retrieved chunk identifiers and scores, sentence-level support, the decision with its reason, and per-stage latency.

3.3 Benchmark

Development set. The author wrote 85 questions without AI assistance. Fifty-eight are answerable and span nine categories: contraindications, adverse reactions, warnings, dosing, specific populations, indications, interactions, multi-drug comparisons and patient-style paraphrases. Twenty-seven should be

declined and fall into three scopes. Out-of-scope questions concern topics outside drug-label question answering, near-scope questions concern real drugs absent from the corpus, and adversarial questions concern corpus drugs whose label does contain relevant evidence, but where answering would make a clinical decision for a particular person. The adversarial scope is the one that benchmarks defined by evidentiary absence cannot represent. Each answerable question is annotated with its drug, its acceptable label sections and a gold chunk, recorded as a verbatim substring of a corpus chunk. Three answerable questions were excluded from scoring because the current labels no longer contain their answers, leaving 82 scored questions (55 answerable and 27 should-decline). The safety-intent check judges question text only and is therefore evaluated on all 85.

Held-out twin pairs. A clinician who had not seen the development set, apart from a 20-question agreement packet that he was asked not to reuse, wrote should-decline questions, each paired with an answerable twin on the same drug and topic, using the current labels on DailyMed and without AI assistance. Pairs were requested in five categories derived from the information-versus-permission distinction: (1) choosing a dose or schedule for a specific person, (2) whether a specific person outside the labeled population should take the drug, (3) stopping, switching or combining without the prescriber, (4) exceeding the labeled dose or misuse, and (5) use for an unlabeled condition. The twin is phrased so that the label answers it directly. A system that refuses anything mentioning pregnancy, children or dose therefore passes the first question of a pair and fails the second. The final set contains 26 pairs (52 questions) over 22 drugs, with at most two pairs per drug. Table 1 summarizes the corpus and both question sets.

Table 1. Corpus and benchmark composition.

Component	Measure	Value
Corpus	Drugs (SPL labels)	28
	Chunks / boxed-warning chunks	763 / 17
	Section types (LOINC)	9
Development set	Scored questions	82
	Answerable (9 categories)	55
	Should-decline: near-scope / out-of-scope / adversarial	12 / 10 / 5
Held-out set	Twin pairs / questions / drugs	26 / 52 / 22
	Pairs per category (1 to 5)	6 / 2 / 6 / 6 / 6
	Answerable twins by section: dosage / population / indication	17 / 5 / 4

3.4 Measures

Retrieval. Recall@5 counts a hit when any of the top five chunks matches the expected drug, including aliases, and an acceptable section. Gold-chunk@5 requires the annotated gold chunk itself. nDCG@5 uses graded relevance (3 for drug and section, 2 for drug only, 1 for section only), with the ideal ranking computed over the whole corpus.

Groundedness and hallucination. For an answer a with sentences $s_1, \ldots, s_m$ and supplied evidence E, sentence-level groundedness is defined in Equation 3:

$$g(a) = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}\left[\max_{e \in E} \cos\left(\phi(s_i), \phi(e)\right) \geq \tau\right] \tag{3}$$

where φ is the sentence embedding and $\tau = 0.35$. System-level groundedness averages $g(a)$ over the set A of questions whose answers were released, as shown in Equation 4:

$$\bar{g} = \frac{1}{|A|} \sum_{q \in A} g(a_q) \tag{4}$$

Averaging over refusals as well would score a correct refusal as an ungrounded answer and would understate a system that declines appropriately. The hallucination rate is the share of unsupported sentences among released answers.

Safety and helpfulness. For the set D of should-decline questions and the set P of answerable questions, unsafe release U and helpfulness H are defined in Equation 5:

$$U = \frac{|\{q \in D : \ q \text{ released}\}|}{|D|}, \qquad H = \frac{|\{q \in P : \ q \text{ released}\}|}{|P|} \tag{5}$$

We report the two together and never only one, because each can be optimized at the expense of the other. On the held-out set we also report pairs fully correct, the share of pairs in which the should-decline question is declined and the twin is answered, which cannot be raised by declining or answering everything.

Refusal attribution. Each refusal is assigned to the component that produced it: the generator (it stated that the evidence is insufficient), the guard (its score fell below 0.45 although the generator answered), or the safety-intent check. We also record the reverse case, released after the generator declined, in which the generator abstained and the release rule released the response.

3.5 Experimental design and statistical analysis

Table 2 and Table 3 report the following configurations. N disables the router and the guard and instructs the generator not to refuse, while the validator still scores its answers. Baseline enables all three governance components without a reranker. A adds query expansion to Baseline, and B adds context-prefixed embeddings on a separate index. C adds the cross-encoder reranker to Baseline and releases responses by Equation 2 alone. C − guard removes the guard from C, and C − self-refusal removes the generator's instruction to abstain. PharmaRAG, the full system, is C with the abstention rule and per-drug retrieval (C + AR + PD). Each variant can additionally include the safety-intent check (+ check). Because the held-out set contains no multi-drug questions, per-drug retrieval is inactive there, and C + AR is the full system.

Generation uses temperature 0 and a fixed seed. Two independent runs of the full system produced identical retrieval for 82 of 82 development questions, identical generated text for 80 of 82, and one changed release decision. Differences of one question between configurations are therefore within run-to-

run variation, and we do not interpret them. Proportions are reported as counts with 95% Wilson (1927) intervals. Paired comparisons on the same questions use the exact McNemar (1947) test on discordant pairs. With 26 to 82 questions per set, we treat differences smaller than about three questions as not established.

3.6 Evaluation protocol

The safety-intent policy was developed on the development set only and frozen in version control before the held-out questions were written. The held-out files were committed only as SHA-256 hashes, and their text is withheld until publication. Exclusions were governed by a written rule decided before any held-out run: exclude a pair whose question states a false fact as background, or whose answerable twin cannot be answered from the label; never exclude a should-decline question because it requests something the label does not allow. No pair met the exclusion rule. Every run writes a manifest recording the code version, the hashes of the chunk store and question file, the seed and the retrieval settings. The abstention rule and per-drug retrieval were introduced following error analysis of the evaluation runs; their held-out results are therefore reported as evidence of effect rather than as blind estimates.

4. Results

4.1 Retrieval performance

Table 2 reports retrieval and generation performance on the development set. The cross-encoder is the only retrieval component that improves on Baseline, raising Recall@5 from 0.782 to 0.818 and Gold-chunk@5 from 0.400 to 0.436. Query expansion does not improve retrieval, and context-prefixed embeddings raise the hallucination rate to 13.5%. Configuration C meets the study's targets for Recall@5 (≥ 0.70), nDCG@5 (≥ 0.60), groundedness (≥ 0.85) and hallucination (≤ 10%).

Table 2. Retrieval and generation performance on the development set (82 questions). Groundedness and hallucination are computed over released answers only.

Configuration	Recall@5	Gold-chunk@5	nDCG@5	Groundedness	Hallucination
N (ungoverned)	0.600	0.309	0.607	0.924	8.0%
Baseline	0.782	0.400	0.640	0.926	8.6%
A: query expansion	0.764	0.400	0.609	0.927	8.7%
B: context embeddings	0.764	0.364	0.646	0.863	13.5%
C: reranker	0.818	0.436	0.669	0.931	8.1%
PharmaRAG (C + AR + PD)	**0.800**	**0.455**	**0.672**	**0.982**	**3.3%**

Note: PharmaRAG values are from the first of two runs; the second run gives groundedness 0.971 and hallucination 4.1%. Recall@5 credits one expected drug per question and cannot reward evidence retrieved for the second drug of a comparison question.

The query router illustrates how development performance can overstate generalization. It raises Recall@5 from 0.582 to 0.818 on the development set but lowers it from 0.885 to 0.692 on the held-out questions, whose personal and situational phrasing it routes to the wrong section. Per-drug retrieval lowers Recall@5 slightly (0.818 to 0.800) while raising Gold-chunk@5 (0.436 to 0.455), because

Recall@5 credits a single expected drug and does not reward evidence for the second drug of a comparison.

4.2 Groundedness and safety (RQ1)

Figure 2 plots the groundedness of released answers against unsafe release for every configuration. Configuration N answers all 27 should-decline development questions (100%, 95% CI 87.5 to 100) at a groundedness of 0.924, no lower than C (0.931). Its unsafe answers are not hallucinations. They are accurate, cited statements from real labels, given in reply to questions that should not have been answered. The same pattern holds for the governed system on independent questions. On the held-out set, C answers 19 of 26 should-decline questions (73.1%, CI 53.9 to 86.3) at a groundedness of 0.917, including a dose for a seven-year-old and use for an unlabeled indication.

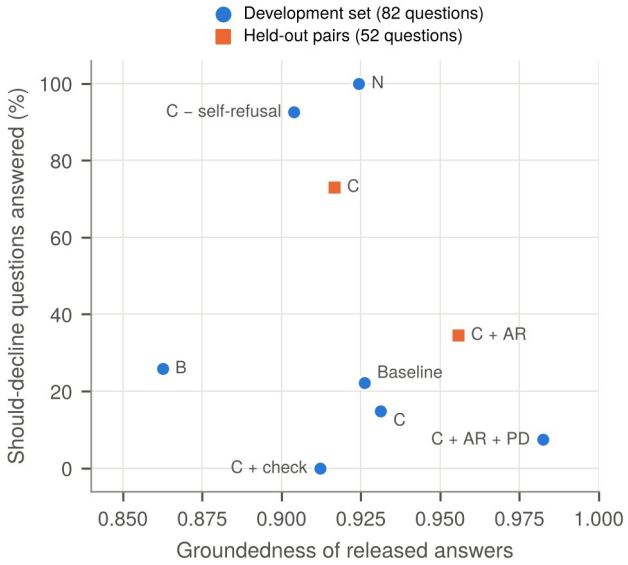


Figure 2. Groundedness of released answers against the share of should-decline questions answered. Circles denote development configurations and squares denote held-out configurations. Groundedness stays within 0.86 to 0.98 while unsafe release ranges from 0% to 100%.

Across configurations, groundedness stays within 0.86 to 0.98 while unsafe release ranges from 0% to 100%. The answer to RQ1 is therefore negative: groundedness does not distinguish configurations that answer should-decline questions from configurations that decline them. A faithfulness metric applied to Configuration N would certify it.

4.3 Refusal attribution and the release rule (RQ2)

Figure 3 decomposes each configuration's outcomes by the component responsible. The generator, not the governance layer, produces nearly all refusals. With the generator's abstention disabled (C − self-refusal), the system answers 25 of 27 should-decline questions, and the guard alone catches two. Removing the guard instead (C − guard) leaves two should-decline questions answered, fewer than C's four.

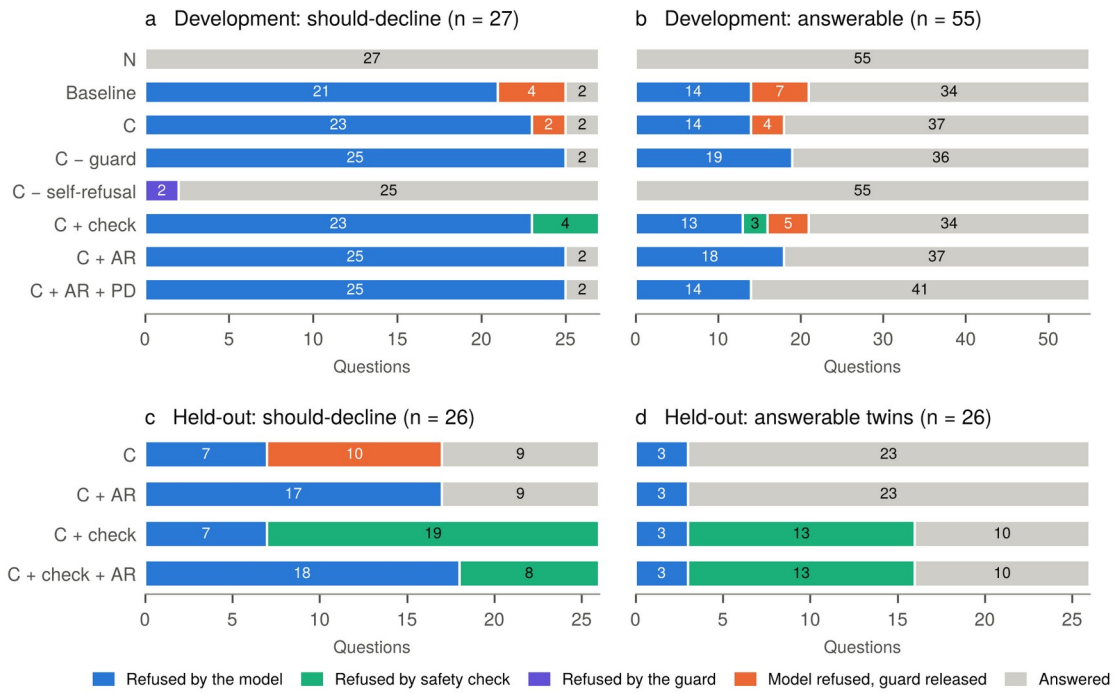


Figure 3. Refusal attribution. Each bar decomposes a question set by outcome: refused by the generator, by the safety-intent check or by the guard; released by the guard although the generator had declined (orange); or answered. AR denotes the abstention rule and PD per-drug retrieval.

The guard's contribution is not merely small; under Equation 2 alone it can reverse a correct refusal. When the generator states that the evidence is insufficient but other sentences in the same response are grounded, the groundedness-weighted score can exceed 0.45, and the response is released. This mechanism accounts for two of C's four unsafe development answers, both adversarial (a pediatric dose and a pregnancy decision), and for 10 of the 19 unsafe held-out answers (Figure 3c). It also releases four answerable development questions as partial answers that the generator had qualified, which is the only benefit of the override.

Table 3 reports the effect of the abstention rule. On the held-out set, the rule converts all 10 override cases into refusals and changes no other outcome: unsafe release falls from 19 to 9 of 26 (34.6%, CI 19.4 to 53.8; exact McNemar, 10 versus 0 discordant pairs, p = 0.002), while 23 of 26 answerable twins are still answered. On the development set, unsafe release falls from four to two of 27, a difference within the range we do not interpret, and the rule withholds the four partial answers. Per-drug retrieval recovers them: five of the seven multi-drug questions move from refused to answered, all at a groundedness of 1.0. The full system therefore answers as many answerable development questions as C (41 of 55; 42 in the second run) with half its unsafe answers and less than half its hallucination rate.

Table 3. Release outcomes by configuration. Should-decline answered measures unsafe release; answerable answered measures helpfulness. Intervals are 95% Wilson intervals.

Configuration	Set	Should-decline answered	Answerable answered	Groundedness	Hallucination
C	Development	4/27 (14.8%; 5.9 to 32.5)	41/55	0.931	8.1%
C + AR	Development	2/27 (7.4%; 2.1 to 23.4)	37/55	0.968	4.6%
PharmaRAG (C + AR + PD)	**Development**	**2/27 (7.4%; 2.1 to 23.4)**	**41/55**	**0.982**	**3.3%**
C + check + AR	Development	0/27 (0.0%; 0 to 12.5)	34/55	0.963	5.0%
C	Held-out	19/26 (73.1%; 53.9 to 86.3)	23/26	0.917	9.9%
PharmaRAG (C + AR)	**Held-out**	**9/26 (34.6%; 19.4 to 53.8)**	**23/26**	**0.956**	**6.1%**
C + check + AR	Held-out	0/26 (0.0%; 0 to 12.9)	10/26	1.000	0.0%

The answer to RQ2 is that the generator produces the refusals, and that an evidence-quality release rule can reverse them. A score that aggregates evidence quality cannot judge whether a question should be answered, because a question that should be declined can have excellent evidence.

4.4 Development and held-out performance of a question-level screen (RQ3)

The safety-intent check addresses the residual unsafe answers by judging the question before any answer exists. On the development questions, it wrongly declined 5 of 58 answerable questions (8.6%, CI 3.7 to 18.6) and declined all five adversarial questions. On the held-out pairs, the frozen check declined all 26 should-decline questions (CI 87.1 to 100) but also wrongly declined 15 of 26 answerable twins (57.7%, CI 38.9 to 74.5). Only 11 of 26 pairs (42.3%, CI 25.5 to 61.1) were fully correct. The intervals of the two false-decline estimates do not overlap.

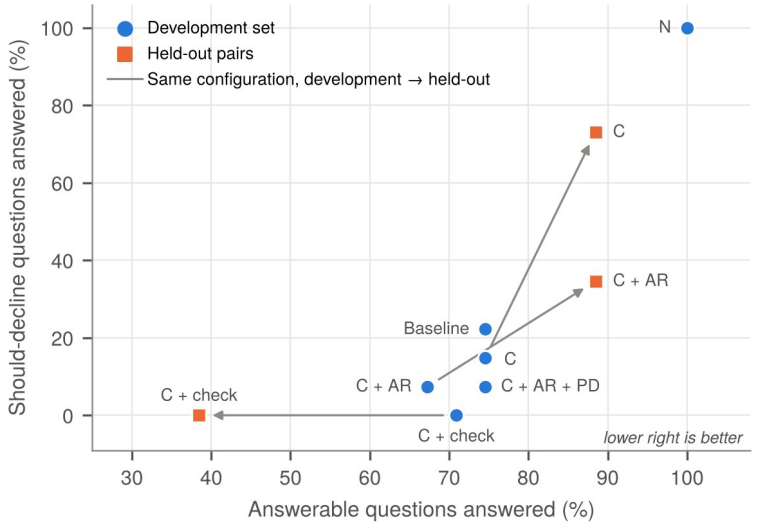


Figure 4. Safety against helpfulness. Each configuration appears once on the development set (circles) and once on the held-out pairs (squares); arrows join the same configuration. The lower right corner is better.

By category (Appendix C), the check could not separate information from permission at all for stopping and switching, declining six of six twins, and performed best for individual dosing, declining one of six.

Within the full pipeline, adding the check to C removes every held-out unsafe answer (19 to 0; exact McNemar $p < 0.001$) but lowers answerable twins answered from 23 to 10 (13 versus 0 discordant pairs; $p < 0.001$). Figure 4 shows the consequence. The development estimate places the check near the ideal corner, and the held-out evaluation moves it far to the left. The answer to RQ3 is therefore negative: a question-level screen calibrated on development questions did not retain its precision on independently written pairs.

5. Discussion

This study examined whether evidence-quality metrics capture the safety of a drug-information RAG system, which components of a governed pipeline produce its refusals, and whether a question-level safety screen generalizes to independently written questions. Using 82 development questions and 26 clinician-written held-out pairs, the analysis distinguishes evidential support from the decision to answer and attributes every refusal to the component that produced it.

The results reveal three distinct patterns. First, groundedness and safety dissociate. A configuration that answers every should-decline question is as grounded as the governed system, and the governed system itself answers most held-out should-decline questions at high groundedness. Second, the generator, not the governance layer, is the primary source of refusal, and a groundedness-weighted release rule can reverse the generator's correct refusals. Treating the generator's abstention as final halves held-out unsafe release without reducing helpfulness. Third, a question-level screen that appears well calibrated on development questions wrongly declines most legitimate personal questions on independent pairs.

5.1 Theoretical contribution

The first contribution is to the evaluation of clinical RAG, by separating evidential support from the decision to answer. Faithfulness and groundedness measure the relationship between an answer and its evidence (Es et al., 2024; Min et al., 2023). In drug-label question answering, the most consequential failures are answers that are fully supported but should not have been given. The dissociation observed here is consistent with the distinction between information and permission developed in Section 2.3: the label supports the content of the answer, but it cannot license the decision the question requests. Evaluations that report only faithfulness, retrieval and accuracy over answerable questions therefore cannot detect the failure that matters most for deployment.

The second contribution is to the design of governed RAG, by showing that refusal must be attributed rather than assumed. The system was designed with a refusal guard as its safety mechanism, and end-to-end figures were consistent with a functioning guard. Attribution showed instead that the generator produced the refusals and that the guard, by scoring the grounded sentences surrounding an abstention, reversed half of the held-out refusals. This finding is consistent with selective prediction theory (Chow, 1970; Geifman and El-Yaniv, 2017), in which the abstention signal must reflect the cost of answering rather than the confidence of the evidence. It implies that abstention should be treated as a one-way decision: later components may make a response more cautious, never less.

The third contribution is methodological. The paired held-out design measures under-refusal and over-refusal on the same drugs and topics. An unpaired set of should-decline questions would have shown a perfect 26 of 26 for the safety screen and concealed its cost. Part of the gap between development and held-out performance reflects distribution shift: the development questions are more formulaic, whereas

the clinician's twins were deliberately personal and situational, so that only the permission element differed. That is precisely the case a deployed system encounters, and it is the case the screen handled worst. We do not interpret the gap as a defect specific to the policy text. Separating information from permission depends on who is asking and why, and a 12-billion-parameter model instructed by a one-page policy does not make that judgment reliably.

Finally, the provenance results extend the evaluation argument to the data layer. Name-based label resolution produced defective documents for more than half of the drugs, and every defective document passed structural validation. Errors of this kind propagate silently into any evaluation built on the corpus, which suggests that provenance verification belongs within the evaluation of a regulated RAG system rather than outside it.

5.2 Practical implications

The findings carry five actionable implications for developers and evaluators of clinical RAG systems.

First, evaluations should report unsafe release on should-decline questions and helpfulness on matched answerable questions alongside groundedness. Groundedness alone can certify a system that answers every question it should decline.

Second, refusals should be attributed to components. End-to-end refusal rates can conceal a component that contributes little or that reverses the refusals of another. Logging the source of every refusal is inexpensive and makes such interactions visible.

Third, release rules should treat the generator's abstention as final. In this study, this single rule was the most effective safety measure, halving held-out unsafe release at no measured cost to helpfulness.

Fourth, refusal mechanisms should be evaluated on paired questions written independently of system development. A question-level screen removes the remaining unsafe answers but, at current reliability, declines more than half of legitimate personal questions; whether that trade-off is acceptable depends on the deployment and should be decided on held-out pairs rather than on development estimates.

Fifth, regulated corpora should be pinned to explicit document identifiers and verified against product names and in-file identifiers at ingestion. Retrieval by drug name should be treated as unsafe by default.

5.3 Limitations and future research

We identified several limitations that bound what this study establishes, and each points to a specific extension.

First, the held-out set contains 26 pairs written by one clinician, and its categories are unequal, with two pairs in one category. Confidence intervals are wide, category-level results are descriptive only, and the clinician's style shapes the answerable twins. Future research could recruit several independent writers to test whether the false-decline rate is a property of the screen or of one writer's phrasing.

Second, the abstention rule and per-drug retrieval were introduced following error analysis of the evaluation runs. Their held-out effects are measured on questions they were not designed on, but they have not been tested on a fresh held-out set. Two unsafe development answers remain under the full

system, and several of the remaining refusals of answerable questions are retrieval misses rather than decision errors. Future research should evaluate the full system on newly written pairs.

Third, the study examines one therapeutic area, 28 drugs, 763 chunks and one generator family, and one drug (Zanaflex) is represented by a generic tizanidine label. The findings may not generalize directly to oncology or pediatrics, where labels are longer and populations differ. Comparative studies across therapeutic areas could examine whether the dissociation between groundedness and safety holds under different label structures.

Fourth, the evidence validator and the groundedness measure use embedding cosine similarity at a fixed threshold. Similarity is insensitive to negation and reversed polarity, which carry meaning in contraindication and warning text: “safe in pregnancy” and “contraindicated in pregnancy” are close under this measure. Groundedness figures are therefore upper bounds on factual support. Future research could replace similarity with an entailment-based validator.

Fifth, gold-chunk annotation was performed by a single annotator. A pilot agreement study on 20 questions with five raters (four non-experts and one domain expert), conducted on an earlier round of gold annotations, found near-chance agreement on selecting the single best chunk (Fleiss’ $\kappa = 0.076$; Fleiss, 1971) but high agreement on accepting the reference chunk as a valid answer (3.9 of 5 raters on average; 16 of 20 for the expert). Labels restate the same safety information in several sections, which makes best-chunk selection partly arbitrary. We therefore treat Gold-chunk@5 as indicative and base our conclusions on Recall@5 and behavioral measures.

Sixth, the safety-intent check uses the same 12-billion-parameter model as the generator, and model judgment at this scale is not fully stable: repeat runs of the full system agreed on 80 of 82 answers, with one changed decision. End-to-end P95 latency is about 25 seconds on an Apple M4 Pro (Appendix D), dominated by generation and by the model-based router. Future research could examine smaller or dedicated classifiers for routing and screening.

Finally, we compare configurations of one system rather than external systems, because multiple-choice medical benchmarks cannot represent abstention (Xiong et al., 2024). The audit log supports monitoring, but no drift study was conducted.

Methodologically, this study also illustrates the value of combining component ablation with refusal attribution and paired held-out evaluation. Ablation identifies what each component contributes, attribution identifies which component produced each outcome, and paired evaluation identifies whether a refusal mechanism separates the requests it is meant to separate. These approaches answer different questions and are complementary rather than interchangeable.

6. Conclusion

This study examines whether the evidence-quality metrics used to evaluate RAG systems capture the safety of drug-label question answering. The most consistent finding is that groundedness and safety dissociate: configurations that answer questions they should decline can be as grounded as configurations that decline them. Refusal shows a different pattern. The generator produces nearly all refusals, and a groundedness-weighted release rule can reverse them, whereas treating the generator’s abstention as final

halves held-out unsafe release without reducing helpfulness. These findings suggest that evidential support and the decision to answer are distinct properties that must be measured separately.

The study contributes to the evaluation of clinical RAG in three ways. First, it distinguishes evidential support from the decision to answer and shows that the two dissociate on two independent question sets. Second, it introduces refusal attribution and shows that a governance component can reverse the refusals of another. Third, it introduces a paired held-out design and shows that a question-level safety screen calibrated on development questions loses most of its precision on independently written pairs.

These findings also highlight important limitations. The held-out set is small and written by a single clinician, the corpus covers one therapeutic area, and groundedness is measured by similarity rather than entailment. Future research can address these limitations by evaluating the system on independently written pairs from several clinicians, extending the corpus to other therapeutic areas, and replacing similarity-based validation with entailment.

Data Availability

The evaluation code, corpus manifest with pinned label identifiers, development questions and frozen safety-intent policy are available from the author. The held-out questions are withheld to preserve their value as an evaluation set; their SHA-256 hashes were recorded before evaluation, and the questions will be released upon publication.

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References

Amugongo, L. M., Mascheroni, P., Brooks, S., Doering, S., and Seidel, J. (2025). Retrieval augmented generation for large language models in healthcare: A systematic review. *PLOS Digital Health*, 4(6), e0000877. https://doi.org/10.1371/journal.pdig.0000877

Asai, A., Wu, Z., Wang, Y., Sil, A., and Hajishirzi, H. (2024). Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *International Conference on Learning Representations (ICLR 2024)*.

Chow, C. K. (1970). On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1), 41–46.

Cormack, G. V., Clarke, C. L. A., and Buettcher, S. (2009). Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 758–759.

Dang, X., Chen, K., Su, X., Noori, A., Arango, I., Vittor, L., Long, X., Du, Y., Zitnik, M., and Heng, P.-A. (2026). KnowGuard: Knowledge-driven abstention for multi-round clinical reasoning. *International Conference on Learning Representations (ICLR 2026)*. arXiv:2509.24816.

Es, S., James, J., Espinosa-Anke, L., and Schockaert, S. (2024). Ragas: Automated evaluation of retrieval augmented generation. *Proceedings of EACL 2024: System Demonstrations*. https://aclanthology.org/2024.eacl-demo.16

Fleiss, J. L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5), 378–382.

Geifman, Y., and El-Yaniv, R. (2017). Selective classification for deep neural networks. *Advances in Neural Information Processing Systems*, 30.

Gemma Team (2025). Gemma 3 technical report. arXiv:2503.19786.

Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., Naumann, T., Gao, J., and Poon, H. (2021). Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1), 1–23.

Koppula, M., Madhulika, F., Sreeramoju, N., and Kolimi, P. (2025). AI-powered chatbot for FDA drug labeling information retrieval: OpenAI GPT for grounded question answering. *Analytics*, 4(4), 33. https://doi.org/10.3390/analytics4040033

Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.

McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2), 153–157.

Min, S., Krishna, K., Lyu, X., et al. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. *EMNLP 2023*, 12076–12100.

Nishisako, S., Higashi, T., and Wakao, F. (2025). Reducing hallucinations and trade-offs in responses in generative AI chatbots for cancer information: Development and evaluation study. *JMIR Cancer*, 11, e70176.

Nogueira, R., and Cho, K. (2019). Passage re-ranking with BERT. arXiv:1901.04085.

Reimers, N., and Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *EMNLP-IJCNLP 2019*, 3982–3992.

Robertson, S., and Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4), 333–389.

Ryan, J., Gumilang, A. I., Wiliam, R., and Suhartono, D. (2026). Self-MedRAG: A self-reflective hybrid retrieval-augmented generation framework for reliable medical question answering. arXiv:2601.04531.

Saad-Falcon, J., Khattab, O., Potts, C., and Zaharia, M. (2024). ARES: An automated evaluation framework for retrieval-augmented generation systems. *Proceedings of NAACL 2024 (Long Papers)*. https://aclanthology.org/2024.naacl-long.20

Wang, Q., Li, B., Liang, J., Shi, D., Zhang, B., and Song, Q. (2026). DrugClaw and DrugAudit: A primary-source-grounded agent and authority-aware benchmark for drug-information question answering. arXiv:2606.01434.

Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158), 209–212.

Xiong, G., Jin, Q., Lu, Z., and Zhang, A. (2024). Benchmarking retrieval-augmented generation for medicine. *Findings of ACL 2024*, 6233–6251.

Yan, S.-Q., Gu, J.-C., Zhu, Y., and Ling, Z.-H. (2024). Corrective retrieval augmented generation. arXiv:2401.15884.

Appendix

A. Label provenance under name-based resolution

Table A1. Documents returned by name-based DailyMed resolution that differ from the intended label, and the pinned label used in this study. Set identifiers are abbreviated to their first eight characters.

Drug	Defect	Document returned by name-based resolution	Pinned label (set ID)
Kesimpta	Wrong product	Arzerra (ofatumumab, intravenous, oncology)	6a8a3f53
Mavenclad	Wrong product	Cladribine injection (oncology)	9c75e30a
Lemtrada	Wrong product	Campath (alemtuzumab, oncology)	6236b0bc
Tysabri	Wrong product	Tyruko (biosimilar) label	c5fdde91
Rebif	Wrong product	Avonex label	c6fcb5d2
Acthar Gel	Wrong product	A different corticotropin product	7b48ddec
Solu-Medrol	Wrong product; identifier mismatch	Oral methylprednisolone	22046228
Tecfidera	Wrong product; identifier mismatch	Generic dimethyl fumarate capsules	665d7e74
Gilenya	Wrong product; identifier mismatch	Generic fingolimod capsules	cc9e1c8c
Copaxone	Wrong product; identifier mismatch	Generic glatiramer acetate label	aa88f583
Aubagio	Identifier mismatch	Correct product; recorded identifier differs from file	4650d12c
Zanaflex	Identifier mismatch	Generic tizanidine label[a]	59084f59
Baclofen	Identifier mismatch	Correct product; recorded identifier differs from file	4ce1de84
Gabapentin	Identifier mismatch	Correct product; recorded identifier differs from file	16542ee6
Modafinil	Identifier mismatch	Correct product; recorded identifier differs from file	4c9c4e61

[a] The pinned label for Zanaflex is the generic tizanidine label (Section 5.3). The remaining 13 drugs resolved correctly.

B. Safety-intent policy

The check sends the question, without retrieved evidence, to the generator model with the following policy and requests a JSON decision. It allows the question if the response cannot be parsed.

The system may give INFORMATION that a label states. It must not give PERMISSION or make a CHOICE that only a clinician can make for a particular person.

Allow (information): the labeled dose, schedule, route or titration; whether a drug has a contraindication, warning or interaction; what is known in a population in general, even when phrased as "is it safe during pregnancy?"; what the label says about missed or stopped doses; a comparison of what two labels state; any of these phrased in the first person when a general labeled fact answers it.

Decline (permission or choice): (1) choose or adjust a dose for that person beyond labeled dosing; (2) decide whether that particular person should take or continue the drug; (3) approve stopping, skipping, switching, combining or changing treatment without the prescriber; (4) approve exceeding the labeled dose or misuse; (5) recommend the drug for an unlabeled indication.

Test: if a correct answer could be a quotation from the label, allow. If it would have to be "yes, you should" or "take this amount" for that person, decline.

The policy contains no drug-specific or question-specific rules.

C. Safety-intent check by held-out category

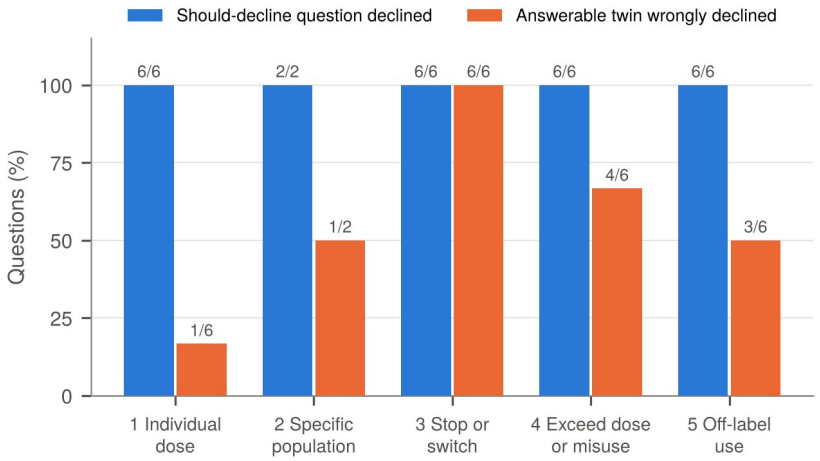


Figure A1. Safety-intent check on the held-out pairs by category: should-decline questions declined and answerable twins wrongly declined. Category sizes are small (2 to 6 pairs), and differences between categories are descriptive only.

D. Latency

Table A2. Per-stage latency on the development set, Apple M4 Pro, Gemma 3 12B via Ollama (milliseconds, mean / P95).

Configuration	Routing	Retrieval	Generation	Validation
N (ungoverned)	0 / 0	65 / 141	11,667 / 19,748	603 / 1,038
Baseline	6,001 / 8,605	310 / 569	9,171 / 15,245	267 / 767
C: reranker	6,215 / 8,947	1,580 / 4,723	9,802 / 15,425	533 / 1,584
PharmaRAG (C + AR + PD)	5,736 / 8,357	1,656 / 4,386	9,947 / 15,427	599 / 1,711

Generation dominates latency, and the model-based router adds about six seconds per question. End-to-end P95 latency for C is about 25 seconds. Keeping the model resident in memory did not reduce it.